

FRANKSTON, Australia

WMO: 948710

Lat: 38.1481S

Lon: 145.1156E

Elev: 20

StdP: 14.69

Time Zone: 10.00 (AUS)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.2	44.0	34.7	29.5	51.8	36.4	31.6	50.7	37.0	52.7	31.6	52.0	4.6	110	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.9	87.9	66.2	84.2	66.0	80.7	65.2	70.0	80.0	68.4	78.6	67.0	76.9	7.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	100.0	73.0	65.1	92.8	72.3	63.3	87.1	71.4	34.0	80.3	32.7	78.7	31.5	77.2	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.6	26.7	23.5	DB	38.3	97.0	1.6	4.3	37.2	100.1	36.3	102.7	35.4	105.1	34.3	108.2	(4)
(5)			WB	37.2	72.9	1.6	1.9	36.0	74.2	35.1	75.3	34.2	76.3	33.1	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	67.3	67.9	65.1	59.8	56.0	52.1	50.9	51.9	54.6	57.2	61.1	64.1	(6)	
	DBStd	7.60	6.25	5.95	5.51	4.46	3.73	3.29	2.89	3.33	4.09	5.00	6.09	6.13	(7)	
	HDD50	58	0	0	0	0	1	15	23	15	3	1	0	0	(8)	
	HDD65	2625	43	32	68	166	278	386	437	406	313	250	153	93	(9)	
	CDD50	3325	536	501	467	295	188	79	51	74	141	223	332	438	(10)	
	CDD65	418	114	113	70	11	0	0	0	0	1	8	35	66	(11)	
	CDH74	2897	859	743	422	32	0	0	0	0	3	40	254	544	(12)	
	CDH80	912	323	236	107	3	0	0	0	0	0	4	60	179	(13)	
(14) Wind	WSAvg	9.8	9.4	8.6	8.6	8.4	9.9	9.7	11.1	10.6	11.0	10.8	9.9	9.9	(14)	
(15) Precipitation	PrecAvg	29.0	1.9	2.1	2.0	2.8	2.3	2.8	2.5	2.5	2.8	2.6	2.6	2.2	(15)	
	PrecMax	39.4	5.2	6.3	3.3	7.1	4.8	5.6	5.1	4.3	5.9	5.3	5.6	5.4	(16)	
	PrecMin	19.3	0.2	0.3	1.1	0.9	0.7	0.7	0.7	0.6	1.0	0.5	0.9	0.0	(17)	
	PrecStd	5.6	1.2	1.5	0.6	1.5	1.1	1.2	1.0	0.9	1.3	1.3	1.2	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.1	91.2	87.8	77.7	69.1	63.9	60.7	66.2	73.0	78.8	86.5	91.4	(19)	
		MCWB	68.5	66.4	65.5	61.8	58.2	56.2	52.3	55.0	57.9	60.6	64.3	65.0	(20)	
	2%	DB	88.0	86.5	82.7	73.6	66.3	60.8	58.3	62.6	68.1	73.7	80.5	84.6	(21)	
		MCWB	67.1	66.7	65.5	60.5	57.1	54.4	51.5	53.6	56.8	59.8	63.4	64.6	(22)	
	5%	DB	83.2	82.4	78.7	70.5	64.4	58.6	56.7	59.6	65.0	70.0	76.2	80.1	(23)	
		MCWB	66.4	66.1	64.6	59.9	56.7	53.1	50.7	52.3	55.3	58.0	62.3	64.3	(24)	
	10%	DB	78.4	78.4	74.7	67.4	62.2	57.2	55.4	57.3	61.9	66.5	72.0	75.3	(25)	
		MCWB	66.0	65.6	63.5	59.1	56.0	52.3	50.3	51.0	53.9	56.7	60.9	63.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.9	71.7	69.9	64.4	61.1	57.9	54.6	57.2	61.0	63.5	68.2	69.6	(27)	
		MCDB	81.5	80.9	80.4	72.1	64.5	61.4	57.7	63.2	68.1	72.5	78.4	80.3	(28)	
	2%	WB	70.3	69.8	67.8	62.6	59.1	55.4	52.9	54.7	58.0	61.1	65.5	67.4	(29)	
		MCDB	80.7	78.5	77.0	68.5	63.7	59.0	56.3	60.7	66.1	69.9	75.3	78.6	(30)	
	5%	WB	68.4	68.3	66.1	61.3	57.7	54.1	51.9	53.0	56.2	59.3	63.7	65.7	(31)	
		MCDB	79.6	77.6	74.7	67.7	62.7	57.5	55.2	58.1	63.0	67.9	73.4	76.5	(32)	
	10%	WB	66.6	66.8	64.5	60.0	56.5	53.0	51.0	51.7	54.6	57.5	61.8	63.8	(33)	
		MCDB	76.9	75.9	73.0	66.3	61.4	56.4	54.4	56.0	60.4	65.1	70.8	73.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	13.9	12.5	10.3	8.1	7.4	6.9	7.7	9.3	11.3	12.9	14.0	(35)	
		MCDBR	24.7	22.8	20.8	15.8	11.3	9.0	8.8	12.0	14.8	19.6	22.8	24.9	(36)	
	5% WB	MCWBR	8.9	8.7	8.2	7.6	6.9	5.8	5.6	6.6	7.8	9.2	9.5	9.7	(37)	
		MCWBR	19.5	17.8	16.4	12.7	9.1	7.5	6.7	9.9	13.1	16.5	18.4	20.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.357	0.359	0.336	0.332	0.319	0.306	0.309	0.321	0.342	0.343	0.350	0.346	(40)	
		taud	2.503	2.508	2.575	2.556	2.575	2.599	2.580	2.540	2.476	2.492	2.500	2.529	(41)	
	Ebn at Noon	Ebn at Noon	309	301	296	276	261	256	262	276	288	303	310	314	(42)	
		Edh at Noon	36	34	29	26	22	20	22	26	32	34	36	35	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2247	1963	1553	1079	712	559	627	886	1269	1701	1998	2233	(44)	
	RadStd	RadStd	136	99	87	59	39	43	31	53	84	127	113	126	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	-151	+254	N/A	(46)
(47)	Regional (36 neighbors)	+0.92	N/A	N/A	+1.51	N/A	N/A	-32	-234	+313	+104	(47)

Nomenclature: See separate page